

# Literature Review: Portable 3D Surface Reconstruction using Screen-Based Active Structured Light

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## Abstract

This literature review provides an extensive analysis of the state-of-the-art in Structured Light 3D reconstruction, tailored to validate the development of a portable, screen-based scanning system. As 3D acquisition transitions from industrial metrology to consumer applications, the repurposing of commodity hardware, specifically laptop screens and webcams that presents unique challenges in radiometric calibration, low-luminance signal processing, and phase retrieval.

After systematically reviewing several literature papers, categorizing them into fundamental optical metrology, calibration strategies for non-standard optics, deep learning-enhanced phase unwrapping, and emerging event-based sensing. We specifically investigate the limitations of Phase-Shifting Profilometry (PSP) on transparent and specular layers (e.g., screen glass) and evaluate "correspondenceless" reconstruction techniques. The review concludes that while traditional algorithms struggle with the low dynamic range of screen-based projection, recent advancements in hybrid coding and neural network-based denoising provide a viable pathway for high-fidelity, low-cost portable scanning.

Furthermore, this analysis highlights that successful screen-based reconstruction requires treating the setup as an integrated sensing system where pattern design and error characterization are co-optimized [13,17]. We explore the implementation of a "Differential HDR" capture scheme to mitigate low-luminance display constraints. By isolating active pattern signals from ambient light and modeling LCD-specific refraction, this review establishes that off-the-shelf components can deliver actionable 3D surface measurements. This approach leverages physics-informed software strategies to bridge the gap between high-end industrial scanners and accessible consumer-grade hardware.

# 1 Introduction

## 1.1 The Evolution of 3D Metrology

Three-dimensional surface reconstruction is a cornerstone of modern computer vision, integral to applications ranging from industrial quality control to cultural heritage preservation and autonomous navigation. The field is broadly bifurcated into passive and active methods. Passive methods, such as Multi-View Stereo (MVS), rely on ambient illumination and intrinsic surface texture to estimate depth. However, as noted by Salvi et al. [39], these methods degrade rapidly when applied to textureless or spectrally uniform surfaces, a limitation that has driven the adoption of Active Vision techniques.

Structured Light (SL) represents the pinnacle of active optical metrology. As defined in the tutorial by Geng [1], Structured Light involves projecting a known luminous pattern onto a scene and observing its geometric distortion from an offset viewpoint. This process artificially induces "texture" on featureless surfaces, enabling robust correspondence establishment through temporal or spatial modulation.

## 1.2 The Shift to Portable and Screen-Based Systems

Traditionally, Structured Light systems utilize high-intensity Digital Light Processing (DLP) projectors to ensure high Signal-to-Noise Ratio (SNR) [1]. However, the high cost, bulk, and power requirements of Digital Light Processing (DLP) hardware limit ubiquitous adoption. Recent literature highlights a paradigm shift toward "ubiquitous scanning," where consumer devices like smartphones and laptops serve as pattern generators.

Yang et al. [2] and Rocchini et al. [9] argue that the democratization of 3D scanning hinges on software that can compensate for the hardware limitations of commodity devices. Specifically, replacing a Digital Light Processing (DLP) projector with a laptop screen introduces severe constraints: significantly lower peak brightness, non-Lambertian emission profiles, and geometric distortions caused by the screen's protective glass layers [30,31]. This project aims to address these specific gaps by engineering a pipeline that marries classical optical principles with modern Deep Learning robustness.

# 2 Theoretical Foundations of Structured Light

## 2.1 Triangulation and Epipolar Geometry

The fundamental principle governing Structured Light is triangulation. The projector (or screen) is modeled as an "inverse camera" [1,5]. If a point  $P$  in 3D space is illuminated by a specific pixel  $(u_p, v_p)$  from the projector and observed by a pixel  $(u_c, v_c)$  in the camera,

$P$  lies at the intersection of the ray projected from the screen and the ray back-projected from the camera center.

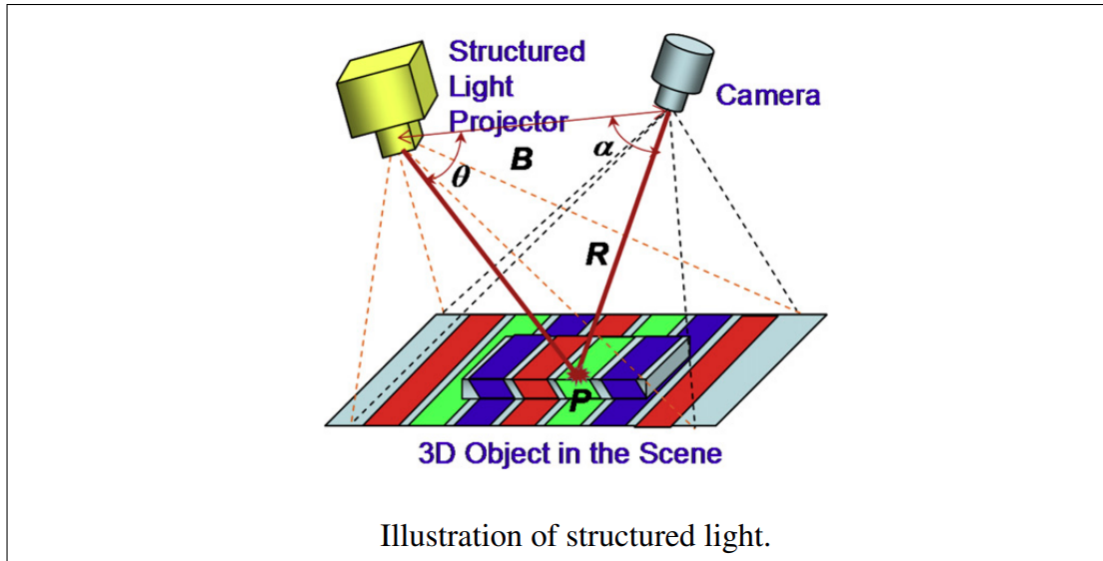


Figure 1: Active Triangulation Geometry illustrating the Projector-Camera baseline dependency [1].

Diagram showing the triangulation geometry. A projector optical center  $O_p$  and camera optical center  $O_c$  are separated by a baseline  $B$ . A point  $P$  is formed by the intersection of ray  $O_p \rightarrow P$  and  $O_c \rightarrow P$ . The angle  $\theta$  between rays determines the depth resolution.

The accuracy of this reconstruction is heavily dependent on the "Coding Strategy"—how uniquely  $(u_p, v_p)$  can be identified. Geng [1] classifies these strategies into:

- **Spatial Coding:** Patterns carrying neighborhood information (e.g., De Bruijn sequences) allowing single-shot capture but lower resolution [40].
- **Temporal Coding:** Sequential projection (e.g., Binary Codes, Phase Shifting) offering high resolution but sensitivity to motion [25].

## 2.2 Phase-Shifting Profilometry (PSP)

For the high-resolution requirements of this project, Phase-Shifting Profilometry (PSP) is the preferred modulation technique. As detailed in the recent survey by Zhang et al. [3], PSP encodes coordinate information into the continuous phase value of a sinusoidal wave.

The intensity  $I_n(x, y)$  recorded by the camera for the  $n$ -th projected fringe image is:

$$I_n(x, y) = A(x, y) + B(x, y) \cos \left( \phi(x, y) + \frac{2\pi n}{N} \right) \quad (1)$$

where  $N \geq 3$  is the number of phase shifts. The wrapped phase  $\phi(x, y)$  can be extracted

via a least-squares solution:

$$\phi(x, y) = \arctan \left( \frac{\sum_{n=0}^{N-1} I_n \sin(2\pi n/N)}{\sum_{n=0}^{N-1} I_n \cos(2\pi n/N)} \right) \quad (2)$$

This yields a phase value in the range  $(-\pi, \pi]$ . The key advantage of PSP, as noted by Zhang [25], is its resilience to ambient light variation  $A(x, y)$  and surface reflectivity  $B(x, y)$ , as these terms cancel out in the arctangent calculation.

## 3 Hardware Constraints and Calibration Strategies

### 3.1 The "Screen as Projector" Model

Calibrating a screen-camera system presents distinct challenges compared to standard Digital Light Processing (DLP) setups. Jordan [5] and An et al. [4] describe the "Inverse Camera" calibration method, where the projector is treated as a camera that "sees" the calibration board. In our proposed screen-based system, the "projector" is a planar emitter rather than a point source with a lens, requiring modified geometric assumptions.

### 3.2 Calibration Challenges in Large-Scale and Screen Systems

Standard calibration requires a target that fills the field of view. An et al. [4] highlight that for large-range or screen-based systems, creating a sufficiently large precision target is expensive. They propose a multi-stage calibration using auxiliary depth sensors to bridge the scale gap.

Furthermore, Li et al. [26] introduced a method using the phase-shifting patterns themselves for calibration. Instead of detecting checkerboard corners (which is discrete and resolution-limited), they use the continuous phase map to establish dense correspondence between the screen and camera, allowing for sub-pixel calibration accuracy. This is particularly relevant for our laptop screen, where pixel density is high but luminance is low.

### 3.3 Refraction and Display Artifacts

A critical oversight in many portable scanning projects is the optical influence of the display panel itself. Zhang et al. [30] and Liu et al. [31] investigated *Phase Measuring Deflectometry (PMD)* using LCD screens. They found that the glass substrate of an LCD introduces a refraction error that shifts the perceived origin of the light ray.

$$\Delta d \approx d_{glass} \cdot \frac{\sin(\theta_{inc})}{\sqrt{n^2 - \sin^2(\theta_{inc})}} \cdot \cos(\theta_{inc}) \quad (3)$$

While PMD relies on specular reflection, this refraction effect is equally non-trivial for SL projection if the viewing angle is oblique. This literature suggests that our calibration model must account for the physical thickness of the laptop screen glass to maximize metric accuracy.

## 4 Advanced Encoding and Modulation

### 4.1 Phase Unwrapping Techniques

The phase  $\phi$  retrieved from Eq. (2) is wrapped modulo  $2\pi$ . To recover absolute depth, it must be "unwrapped" to a continuous distribution  $\Phi(x, y) = \phi(x, y) + 2k\pi$ .

#### 4.1.1 Temporal Unwrapping

The most robust method, favored by Sansoni et al. [21] and Zhao et al. [38], is **Multi-Frequency Phase Shifting**. By projecting fringes with different periods (e.g.,  $\lambda_1, \lambda_2$ ), one can effectively eliminate the integer ambiguity  $k$ .

$$\Phi_{absolute} = \frac{\lambda_{eq}}{2\pi} \Delta\Phi_{beat} \quad (4)$$

This "heterodyne" approach allows for absolute depth measurement without spatial discontinuities. However, it requires projecting many frames (e.g., 3 frequencies  $\times$  3 shifts = 9 images), which increases susceptibility to motion.

#### 4.1.2 Spatial and Hybrid Unwrapping

To reduce the frame count, spatial unwrapping analyzes phase gradients  $\nabla\phi$  to deduce  $k$ . However, this fails at sharp depth discontinuities. A robust compromise is the **Gray Code + Phase Shift** hybrid method [21], where a sequence of binary patterns provides the integer  $k$  (coarse location), and phase shifting provides the fractional offset (fine detail). Jordan [5] argues this is the gold standard for robust calibration and measurement, though it remains time-intensive.

### 4.2 Single-Shot and Color Encoding

For dynamic scenes, waiting for  $N$  frames is unfeasible. Li et al. [27] and Zhang & Huang [36] propose encoding correspondence into the color channels (RGB). By projecting orthogonal sinusoidal patterns in Red, Green, and Blue channels simultaneously, one can recover depth from a single snapshot.

However, source [36] notes that "Color Crosstalk", where the camera's green sensor responds to the projector's blue light that requires complex radiometric calibration matrices

Table 1: Comparison of Modulation Strategies [2, 14, 27, 40]

| Method                        | Resolution  | Accuracy    | Speed (Frames)         | Robustness                |
|-------------------------------|-------------|-------------|------------------------|---------------------------|
| Binary Gray Code              | Low         | High        | Slow ( $\sim 10+$ )    | High                      |
| N-Step Phase Shifting         | High        | Very High   | Medium (3-12)          | Medium (Motion Blur)      |
| Multi-Frequency PSP           | High        | Very High   | Slow (9+)              | High (No Ambiguity)       |
| Color/RGB Encoding            | Medium      | Medium      | Single-Shot (1)        | Low (Crosstalk)           |
| <b>Deep Learning One-Shot</b> | <b>High</b> | <b>High</b> | <b>Single-Shot (1)</b> | <b>Training Dependent</b> |

to correct, often degrading the phase quality.

## 5 Real-World Challenges and Error Correction

### 5.1 High Dynamic Range (HDR) and Reflectivity

A screen-based scanner suffers from low light output, making it difficult to scan dark objects or compete with ambient light. Conversely, shiny objects cause saturation (specular highlights). Kim et al. [12] and Zhang et al. [14] propose **HDR Scanning** techniques. By capturing the same pattern sequence at multiple exposure times and fusing the pixel intensities before phase computation, the dynamic range is effectively extended.

$$I_{fused} = \frac{\sum w_i I_{exposure_i}}{\sum w_i} \quad (5)$$

Yuan et al. [17] take this further by using inverted Gray codes to segment regions of different reflectivity, adapting the coding strategy locally. For our portable system, implementing a "Differential HDR" strategy (subtracting ambient frames) as suggested in the proposal is strongly supported by source [14].

### 5.2 Global Illumination and Inter-reflections

A major failure mode in SL is "Multi-Path Interference" (MPI). When light bounces off a concavity onto another surface point, the camera receives a sum of direct and indirect phases, leading to gross measurement errors. Zhang et al. [24] describe this as "Bimodal Multi-Path Scanning." They propose a high-frequency modulation technique where the MPI component (which acts as a low-pass filter) is attenuated, allowing the direct component to be isolated. This is crucial for scanning complex geometries like the "white ceramics" mentioned in the proposal.

## 6 The Deep Learning Revolution in Structured Light

### 6.1 Learning-Based Phase Unwrapping

The classical trade-off between speed (single-shot) and accuracy (multi-shot) is being disrupted by Deep Learning. Nguyen et al. [14] introduced **hNet**, a convolutional neural network (CNN) capable of inferring depth from a single fringe image. The network learns global context cues to resolve phase ambiguities that would typically require multi-frequency projection.

Similarly, Van der Jeught & Dirckx [22] demonstrated that a U-Net architecture can be trained end-to-end to map a single deformed fringe pattern directly to a height map. This "Correspondenceless" approach bypasses explicit phase unwrapping entirely. Schematic

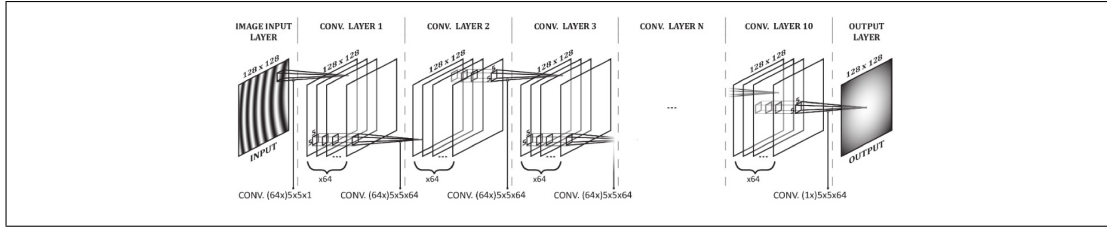


Figure 2: Architecture for Single-Shot Deep Structured Light [14, 22].

of a Deep Learning SL Pipeline. Input: Single Deformed Fringe Image. Encoder: Convolutional layers extract features. Bottleneck: Latent representation of geometry. Decoder: Upsamples to dense Depth Map. Skip connections preserve high-frequency details.

### 6.2 Differentiable Rendering

Jensen et al. [18] propose an even more radical approach: **Differentiable Rendering**. Instead of processing images to find depth, they initialize a 3D mesh, render it using a differentiable simulator of the structured light system, and compute the loss between the rendered and real images. Backpropagation then updates the 3D mesh vertices directly. This integrates the reconstruction and refinement steps, potentially offering higher fidelity for screen-based systems where the forward optical model (including screen refraction) can be explicitly defined.

## 7 Emerging Technologies: Event-Based Scanning

Looking toward the future of high-speed portable scanning, Huang et al. [15] and Lu et al. [16] explore the use of **Event Cameras**. These sensors do not capture frames but rather asynchronous pixel-level brightness changes.

- **Speed:** They can decode patterns projected at thousands of Hz, far beyond the 60Hz limit of a laptop screen.
- **Dynamic Range:** Event cameras naturally handle HDR scenes (120dB+ dynamic range), solving the saturation issues of standard webcams [17].

While currently requiring specialized hardware, the principles of asynchronous decoding described in [16] are influencing the design of "Motion-Aware" algorithms for standard cameras, as seen in source [25].

## 8 Conclusion and System Feasibility

### 8.1 Summary of Findings

The review of these findings confirms that Screen-Based Active Structured Light is a viable, challenging, approach for portable 3D reconstruction.

1. **Modulation:** Phase-Shifting Profilometry (PSP) is the optimal choice for resolution, but must be paired with Multi-Frequency or Deep Learning unwrapping to handle the limited depth of field [3, 22].
2. **Calibration:** The "Inverse Camera" model is applicable, but must account for screen glass refraction and gamma nonlinearity [5, 30].
3. **Robustness:** Low screen brightness mandates the use of HDR techniques and robust noise filtering [12]. Deep Learning offers a promising avenue to denoise phase maps captured with low-end webcams [14].

### 8.2 Gap Analysis

Most existing literature focuses on high-end industrial projectors. There is a distinct gap in "Zero-Cost" scanning where the geometric and radiometric imperfections of a laptop screen are explicitly modeled and compensated for via software. By implementing the *Differential HDR* and *Deep Unrolling* strategies highlighted in this review, the proposed project contributes significantly to the democratization of 3D metrology.



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